

RevInsight Take-Home — Part 1

Should the store replace the departed salesperson, and when?

Assumptions

The dataset does not explicitly define several business details. To complete the analysis, the following assumptions were made:

- The “Sales” values, salary values, and commission values are all expressed in the same monetary units. Since the exact currency scale is not provided, all values are treated as consistent relative units throughout the analysis.
- A salesperson whose sales drop to zero and remain at zero permanently is assumed to have left the business rather than being temporarily absent.
- Historical patterns of employee productivity and store seasonality are assumed to remain broadly stable over the next six months. No major changes in store operations, mall traffic, pricing strategy, or market conditions are assumed.
- New hires are assumed to follow a productivity ramp similar to historical hires observed in the dataset.
- Existing employees are assumed to continue performing in line with their recent productivity trends.
- Adding a new salesperson does not materially reduce the productivity of existing salespeople (i.e., no significant sales cannibalization effect).

Recommendation

Based on analysis of 81 months of historical store data, the recommendation is to **hire a replacement salesperson now** rather than continuing to operate with one fewer employee.

Current Situation

At the time salesperson sp9 resigned, the store had 8 active salespeople. Since the resignation, the store has operated with 7 active salespeople, which aligns with the owner's typical target staffing level.

Therefore, the decision is not whether to return to the historical target of 7 employees, but whether **adding an 8th salesperson today** would increase expected profitability over the next six months.

Executive Summary

The store has already been operating without replacing sp9 for approximately five months. While the store remained operational during this period, a forward-looking profit analysis indicates that hiring a replacement now is expected to generate the highest gross margin over the next six months.

Scenario	Expected 6-Month Gross Margin
Remain at current staffing level	~196.6
Hire replacement immediately	~202.7
Delay hiring by 1 month	~200.4
Delay hiring by 2 months	~198.3
Delay hiring by 3 months	~196.4

Hiring immediately is expected to increase six-month gross margin by approximately 6.1 units versus remaining at the current staffing level — roughly a **3% improvement**.

Key Findings

1. The departing employee was a highly productive veteran

Salesperson sp9 worked for approximately 47 months and was generating roughly 20–21 sales units per month before leaving. Historical data shows that new hires begin substantially below this level and require significant time to reach veteran productivity. A replacement therefore should not be viewed as a one-for-one substitution. Instead, the store is replacing a highly productive veteran with a lower-productivity employee who gradually improves over time.

2. New hires ramp slowly

Using historical employee performance data, a productivity ramp curve was fitted for new salespeople. A typical new hire generates approximately 6–8 sales units per month initially and may require several years to reach the productivity levels achieved by long-tenured employees such as sp9. This slow ramp explains why replacing a veteran does not immediately restore lost productivity.

3. Seasonality strongly affects hiring decisions

Store sales exhibit a recurring seasonal pattern, with peak months generating roughly twice the sales of the weakest months. This matters because salesperson profitability depends on sales volume: a new hire starting during a weak seasonal period may initially generate insufficient sales to cover compensation costs, while the same hire can be profitable when starting during strong seasonal months.

4. Breakeven analysis

Using the owner's profit formula:

$$\text{Gross Margin} = 50\% \times \text{Sales} - \text{Salary} - 5\% \times \text{Sales}$$

which simplifies to:

$$\text{Gross Margin} = 45\% \times \text{Sales} - \text{Salary}$$

With a monthly salary cost of 3 units, a salesperson must generate approximately **6.67 sales units per month to break even**. Historical ramp estimates suggest new hires start close to this threshold. Depending on seasonal demand, a new hire may fall below breakeven during the weakest seasonal periods and exceed breakeven comfortably during peak periods.

Why Hiring Immediately Is Best, In This Case

Although hiring timing matters in general, this specific resignation occurs immediately before the strongest part of the store's seasonal cycle. The next several months correspond to historically high-demand periods. As a result, a new hire is expected to exceed breakeven quickly and contribute positive margin during the months when customer traffic is strongest.

When delayed-hiring scenarios were modeled, every delay reduced expected profit because the new employee missed the most favorable seasonal months. Therefore, while automatic replacement of every future resignation is not recommended, hiring immediately is recommended in this particular case.

Recommended Staffing Rule

Replace the current rule:

“Always refill headcount immediately when someone resigns.”

with:

“Estimate the expected incremental gross margin of a replacement over the next six months, taking into account employee ramp-up and seasonal demand. Hire when expected incremental margin is positive, and delay hiring when a replacement would spend most of its early months in a seasonal trough.”

This rule aligns staffing decisions directly with profitability rather than fixed headcount targets.

Model Limitations

This analysis relies on historical patterns observed in the dataset. The productivity ramp curve is estimated from historical employee tenure and performance, while seasonality is estimated from approximately 81 months of sales history. Future results may differ if customer demand, mall traffic, compensation structure, hiring quality, or store operations change materially. Additionally, the model assumes that a new salesperson does not significantly cannibalize sales from existing staff and that future hires resemble historical hires in productivity growth.